

# Martin Nunction

Martin Nunction(a pun for Notation/Function), Is a large number notation created by Martin Wanralfs or UpBeatSquirrel925 in Fandom wiki.

## A.Definitions

### 1.Martin Aunction (MA)

Martin Aunction is the very first function in Martin nunction, written as:

$$A(a,b,c)$$

Let a be any non-zero finite positive integer, a is the argument, a is the first entry.

Let b be any non-zero finite positive integer, b is the amount of (c-th) hyperoperation, b is the second entry.

Let c be any non-zero finite positive integer, c is the hyperoperation level, c is the third entry, where  $c = 1$  is the Knuth up arrows Hyperoperation ( $\uparrow^b$ , b amount)

$$A(a) = a$$

$$A(a,b) = a^b$$

General Martin Aunction expansion:

$$A(a,b,c) = a (c)^b a = a (c)^{b-1} (a (c)^{b-1} (a (c)^{b-1} \dots \#.$$

Right associatively, using right operand a value as the amount of iteration.

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

$$A(a,b,c) = a (c\text{-th level hyperoperation})^b a.$$

General Martin Aunction leveling down:

$$A(a,1,c) = a (c)^1 a = a (c-1)^a a, \text{ or}$$

$$A(c)^1 n = a (c-1)^n a.$$

Where  $b = 1$ , b value replaced by right operand a, while c is subtracted by 1, moving to (c-1)-th hyperoperation.

Examples:

$$A(1,b,c) = 1 (c)^b 1 = 1$$

$$A(1,1,1) = 1 \uparrow^1 1 = 1^1 = 1$$

$$A(1,n,n) = 1 (n)^n 1 = 1$$

Where  $c = 1$  is the Knuth's up arrow hyperoperation ( $\uparrow^b$ , with  $b$  amount),  $a = 1$ , left  $a$  is the argument while right  $a$  is hyperoperation expansion operand, and  $b = 1$  is the amount of current level hyperoperation,  $a$  becomes the result ignoring the second and third entry iff  $a = 1$ .

$$A(2,2,2) = 2 (2)^2 2 = 2 (2)^{12} = 2 \uparrow^2 2 = 2 \uparrow 2 = 2^2 = 4.$$

$$A(2,b,c) = 2 (c)^b 2 = 2 (c)^{b-1} 2 = 2 (c)^{b-2} 2 \dots$$

Operation will terminate to 4 such that  $a = 2$ .

$$A(3,2,1) = 3 \uparrow^2 3 = 3 \uparrow (3 \uparrow 3) = 3 \uparrow 27 = 7,625,597,484,987.$$

## 2. Martin Array Aunction (MAA, Level 1)

The next level in Martin Nunction, extending the array length of Martin Aunction, written as

$$A(a,b,c,d,e,\#)$$

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

Where array entry must be a non-zero finite positive integer or an array.

General 4 Entry Expansion

(note: do 4 entry expansion before pairing rule, such that  $b=d \Rightarrow 2$ ):

$$A(ab,c,d) = A(a,b,(a,b-1,c,d),d-1)$$

General Pairing rule for Entry  $>3$  where there's a number 1 followed by a number 2, and where 1 followed by a number  $>2$ , scanning right to left:

$$A(a,1,2,\#) = A(a,3,1,\#)$$

$$A(a,1,c,\#) = A(a,(a,(\dots\#\dots)),c-1,\#),c-1,\#)$$

Where  $b$  is now replaced with an entire array nested into  $b$  entry  $c$  times, subtracting  $c$  by 1 on the outer array and all inner arrays

General  $n$  entries:

$A(a,b,c,d,\#) \rightarrow A(a,b,(a,b-1,c,d,\#),c-1,d,\#)$ , Entries after  $d$  are for applying pairing rule such that any entry except for  $a = 1$

Example:

$$A(3,2,2,2) = A(3,2,(3,1,2,2),1) \rightarrow A(3,2,(3,1,2,2)), \text{ remove trailing 1, according to rule 3.}$$

$\rightarrow$  Applying pairing rule, for pair  $b=1$  and  $c=2$ ,  $(1,2)$ .

$$A(3,2,(3,3,1,2))$$

$\rightarrow$  Applying pairing rule, for pair  $c=1$  and  $d=2$ ,  $(1,2)$ .

$$A(3,2,(3,3,3,1)) \rightarrow A(3,2,(3,3,3)), \text{ remove trailing 1.}$$

$\rightarrow$  Let  $X = A(3,3,3)$

$$\rightarrow A(3,2,X) = 3 (X)^2 3.$$

Example:

$$A(3,2,3,2) = A(3,2,(3,1,3,2),1)$$

$$\rightarrow A(3,2,(3,1,3,2)) \rightarrow$$

$$A(3,2,(3,(3,(3,1,2,2),2,2),2,2))$$

$$\rightarrow A(3,2,(3,(3,(3,3,1,2),2,2),2,2))$$

$$\rightarrow A(3,2,(3,(3,(3,3,3),2,2),2,2))$$

$$\rightarrow A(3,2,(3,(3,(3,3,3),2,2),2,2)), \text{ call the result of } (3,3,3) = X$$

$\rightarrow A(3,2,(3,(3,X,2,2),2,2))$ , Suppose we run the 4 entry expansion operation X times and evaluate the inner arrays of (3,X,2,2) expansion, call the evaluated Value as X\_1

$\rightarrow A(3,2,(3,X_1,2,2))$ , Suppose we run the 4 entry expansion operation X\_1 times and evaluate the inner arrays of (3,X\_1,2,2) expansion, call the evaluated Value as X\_2.

$$\rightarrow A(3,2,X_2) = 3 (X_2)^2 3.$$

Example:

$$A(3,2,2,2,2) \rightarrow A(3,2,(3,1,2,2,2),1,2)$$

$$\rightarrow A(3,2,(3,3,1,2,2),3,1) \rightarrow A(3,2,(3,3,1,2,2),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,1,2,2),1,2),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,1,1,2,2),1,2),1,2),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,1,3,1,2),3,1),3,1),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,1,3,1,2),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,1,3,3,1),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,1,3,3),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,1,2,3),2,3),2,3),2,3),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,1,3),2,3),2,3),2,3),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,3,1,2),2),2),2),3),2,3),2,3),2,3),3),3),3)$$

$$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,3,3,1),2),2),2),3),2,3),2,3),2,3),3),3),3)$$

$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,3,3),2),2),2),3),2,3),2,3),2,3),3),3),3)$ , Let the evaluated value of innermost array “(3,3,3)” be called X.

$$\rightarrow A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,X,2),2),2),3),2,3),2,3),2,3),3),3),3)$$

→A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,2,X,2),1),2),2),3),2,3),2,3),2,3),3),3),3)

→A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,2,X,2))),2),2),3),2,3),2,3),2,3),3),3),3)

→A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,2,(3,1,X,2),1))),2),2),3),2,3),2,3),2,3),3),3),3)

→A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,2,(3,1,X,2))))),2),2),3),2,3),2,3),2,3),3),3),3), Suppose we run the pairing rule and evaluate inner arrays of (3,1,X,2), Let the result called X\_1

→A(3,2,(3,3,(3,2,(3,(3,(3,(3,3,(3,3,(3,3,(3,2,(3,X\_1,X-1,2))))),2),2),3),2,3),2,3),2,3),3),3),3)

### 3.Martin Array Nunction (MAN, base array universe)

Martin Array Nunction(MAN)

A\_(a)(n,l).

let (a) = array level

let n = value of entries in current array level

let l = length or amount of entries in current array level

A\_(1)(n,4) = A(n,n,n,n)

A\_(2)(n,3) = B(n,n,n)

General Leveling down when l = 1 or 2 or 3:

A\_(n)(n,1) = n(n) = n

→ A\_(n-1)(n,n)

A\_(n)(n,2) = n(n,n) = n^n, let the result be X

→ A\_(n-1)(n,X)

A\_(n)(n,3) = n(n,n,n) = n (n-th hyperoperation)^n n, let the result be X

→ A\_(n-1)(n,X).

Replace parameter l with result of (n) level array calculation

Then subtract (n) by 1.

General l = 4 expansion for any (a):

A\_(a)(n,4) = a(n,n,n,n) = a(n,n,(n,n-1,n,n),n-1)

Array with l > 4 or more than 4 entries are only for applying the pairing rule.

General > 1 entry parameter (a):

$$A_{-}(a,b)(n,l) = \text{Let } X = A_{-}(a)(n,l)$$

$$\rightarrow A_{-}(X,b-1)(n,l)$$

$$A_{-}(a,b,\#,j,k)(n,l) = \text{Let } X = A_{-}(a,b,\#,j)(n,l)n,l$$

$$\rightarrow A_{-}(X,X,\#,X,k-1)(n,l)$$

Calculate (a) parameter with one lower entry length before k-th entry.

Then Replace every entry before k with X and subtract k-th entry by 1.

MAN Parallels:

$$A_{-}[p](a)(n,l)$$

$$A_{-}[1](a)(n,l) = A_{-}(a)(a)(n,l)$$

Where (a) is the first parallel of MAN

Where p is the amount of parallels parameter

Example:

General 1 parallel with single entry:

$$A_{-}[1](n)(n,n) = A_{-}(n)(n,n) = \text{Let } X = \text{Result of } A_{-}(n)(n,n).$$

$$\rightarrow A_{-}(n-1)(X,X,X,\#)(X,X)$$

Calculate no parallel MAN, subtract first parallel first entry by 1, then replace every parameter from no parallel MAN with Calculated value and Calculated result Amount of Entries for MAN, then replace (n,l) parameters with calculated result.

$$A_{-}[1](1)(2,2) = A_{-}(1)(1)(2,2) = \text{Calculate } A_{-}(1)(2,2) \text{ first, } A_{-}(1)(2,2) = A(2,2) = 4.$$

$$\rightarrow A_{-}(4,4,4,4)(4,4).$$

(Note:  $A_{-}[1](a)(n,l) = 1$  Parallel MAN.

Parallels copy the amount of entries and value of MAN (a) parameter)

General 1 parallel with >1 entries:

$$A_{-}[1](a,b,c,\#,j,k)(n,l) = A_{-}(a,b,c,\#,j,k)(a,b,c,\#,j,k)(n,l) = \text{Let } X = \text{result of } A_{-}(a,b,c,\#,j)(a,b,c,\#,j,k)(n,l). \rightarrow A_{-}(X,X,X,\#,X,k-1)(X,X,X,\#,X,X)(X,X).$$

General n parallels with 1 entry:

$$A_{-}[n](n)(n,l) = A_{-}(n)(n) \dots \#(n,l) = \text{Let } X = \text{result of } A_{-}(n) \dots \#(n,l).$$

$$\rightarrow A_{-}(n-1)(X,X,\#) \dots \#(X,X)$$

Calculate [n-1] parallel MAN, subtract n-th parallel first entry by 1, replace every parallels below n-th parallels value and entry amount by calculated value, then replace (n,l) parameter value by calculated value

(note: Octothorpe here means unchanged remainder of parallels or arrays)

General n parallel with >1 entries:

$A_{[n]}(a,b,\#,j,k)(n,l) = A_{(a,b,\#,j,k)}(a,b,\#,j,k) \dots \#(n,l) = \text{Let } X = \text{result of } A_{(a,b,\#,j)}(a,b,\#,j,k) \dots \#(n,l).$

$\rightarrow A_{(X,X,\#,X,k-1)}(X,X,\#,X,X) \dots \#(X,X).$

MAN Parallel Array with 2 entry

$A_{[p,q]}(n)(n,l) = \text{Let } X = \text{Result of } A_{[p]}(n)(n,l).$

$\rightarrow A_{[X,q-1]}(X,X,X,\#)(X,X)$

Calculate MAN Parallel array one entry lower than q, then replace every entry before q with calculated value, subtract q by 1, then replace MAN (a) parameter with Calculated value amount of entries and value, then replace (n,l) parameter with Calculated value.

General Parallel Array with n entries:

$A_{[p,q,\#,y,z]}(n)(n,l) = \text{Let } X = \text{result of } A_{[p,q,\#,y]}(n)(n,l).$

$\rightarrow A_{[X,X,\#,X,z-1]}(X,X,\#)(X,X)$

Calculate MAN Parallel array one entry lower than z-th entry, then replace every entry before z-th entry with calculated value, subtract z-th entry by 1, then replace MAN (a) parameter with Calculated value amount of entries and value, then replace (n,l) parameter with Calculated value.

#### 4.Bangs (!)

An Array operator of MAN, the first priority Operator, written in the operator slots which are second to last entry in MAN p parameter entry.

$A_{[\#,operator,z]}(a)(n,l)$  or  $A_{[p,!,q]}(a)(n,l)$

General entry = 2 with single or more bangs:

$A_{[p,!,q]}(a)(n,l) = A_{[p,p,\#]}(a)(n,l)$

p parameter with  $A_{[p]}(p)(p,p)$  entries.

General MAN a parameter array with n bangs:

$A_{[p,!^n,q]}(a)(n,l) = A_{[p,p,\dots\#,!^{n-1},a]}(a)(n,l)$

MAN parallel array filled with a with  $A_{[p]}(p)(p,p)$  Entries and n-1 bangs.

>2 entry with bangs recursion:

$A_{[p,q,!,r]}(a)(n,l) = \text{Let } X = A_{[p,!,q]}(a)(n,l), \text{ calculate 2 entry p parameter with single bang first.}$

$\rightarrow A_{[X,X,!,r-1]}(X,X,X\#)(X,X).$

$A_{[p,q,r,!,s]}(a)(n,l) = \text{Let } X = A_{[p,q,!,r]}(a)(n,l), \text{ calculate 3 Entry p parameter with single bang first.}$

$\rightarrow A_{[X,X,X,!,s-1]}(X,X,X\#)(X,X).$

General with n bangs:

$A_{[p,q,\#,y,!^n(z)]}(a)(n,l) = \text{Let } X = A_{[p,q,\#,!^n(y)]}(a)(n,l), \text{ calculate p parameter with one entry lower with n bangs first.}$

$\rightarrow A_{[X,X,\#,X,!^n(z-1)]}(X,X,X\#)(X,X)$

Replace every entry before bang with X, along with n and l parameters, with X entries and values in a parameter.

## 5. Tiers (/n/)

The second priority operator.

$A_{[p,/n/,q]}.$

General Tiering:

$/n/ = \text{Tier 1 with value n, } //n// = \text{Tier 2 with value n, (k-th tier}(n) = \text{k-th tier with value n, higher structure, tier, value} = \text{less priority.}$

General tier 1 with  $n = 1$  and  $n > 2$ :

$A_{[p,/1/,q]}(a)(n,l) = \text{Let } X = A_{[p,q]}(a)(n,l), \rightarrow A_{[p,!^n(A_{[p,q]}(a)(n,l))]}(a)(n,l)$

Evaluate X, then subtract tier 1 Value by 1, which becomes X amounts of Bangs, with entire function nested into bang exponent, where innermost nested function =  $A_{[p,!^n(X),q]}(a)(n,l)$

$A_{[p,/n/,q]}(a)(n,l) = \text{Let } X = A_{[p,q]}(a)(n,l)$

$\rightarrow A_{[p,/n-1/,!^n(A_{[p,q]}(a)(n,l))]}(a)(n,l)$

Evaluate X, then subtract tier 1 value by 1, which becomes X amounts of Bangs added to the second rightmost entry as higher priority operator, with entire function nested into it's exponent, Where the innermost nested function =  $A_{[p,/n-1/,!^n(X),q]}(a)(n,l)$

General Tier 2 and Tier k with value n:

$A_{[p,//n//,q]}(a)(n,l) = \text{Let } X = A_{[a,b]}(a)(n,l)$

→  $A_{[p, //n-1//, /A_{\#}/, q]}(a)(n, l)$

Where innermost nested function =  $A_{[p, //n-1//, /X/, q]}(a)(n, l)$

Nest entire function X times inside Tier 1 bracket and subtract Tier 2 value by 1, then add lower tier operator on the right side of the higher tier operator in the operator slots

k-th tier:

$A_{[p, (k\text{-tier}(n)), q]}(a)(n, l) = \text{let } X = A_{[p, q]}(a)(n, l)$

→  $A_{[p, (k\text{-tier}(n-1)), (k-1 \text{ tier}(\#)), q]}(a)(n, l)$

Where innermost nested function =

$A_{[p, (k\text{-tier}(n-1)), (k-1 \text{ tier}(X)), q]}(a)(n, l)$

Nest entire function into (k-1)-th tier bracket and subtract k-th tier value by 1, then add lower tier operator on the right side of the higher tier in the operator slots.

## 6. Martin Kunction (MK)

Index of Structure and tiers, written as:

$M_k(m, n, l)$

Let k = k-th tier structure within Martin Nunction.

Let m = tier within Current tier structure

Let n = amount of current tier within the structure

Let  $l = A_{[l, l, \#]}(l)(l, l)$ , with l entries for p parameter.

$M_k(m, n, 3) = A_{[3, 3, (k\text{-th structure tier } m(n)), 3]}(3)(3, 3)$

Example:

$M_1(1, 1, 3) = A_{[3, 3, /1/, 3]}(3)(3, 3)$

$M_1(2, 1, 3) = A_{[3, 3, //1//, 3]}(3)(3, 3)$

$M_2(1, 2, 2) = A_{[2, \{2\}, 2]}(2)(2, 2)$

General Martin Kunction:

$M_k(m, n, l) = A_{[\#, (k\text{-structure tier } m(n)), l]}(l)(l, l) = \text{let } X = A_{[\#, l]}(l)(l, l)$

→  $A_{[\#, (k\text{-structure tier } m(n-1)), (k\text{-structure tier } m-1(A_{\#}))]}(l)(l, l)$

Calculate X, subtract k-th structure m-th tier value by 1, then add k-th Structure (m-1)-th tier with value X, nest entire function into k-th structure (m-1)-th tier bracket.

Where Innermost nested function =  $A_{\#}(k\text{-structure } m(n-1)), (k\text{-structure tier } m-1(X)), l(l)(l, l)$ .

Leveling Down Structure:

$A_{\#}(k\text{-structure tier } 1(1)), l(l)(l, l) = \text{Let } X = A_{\#}, l(l)(l, l)$

→  $A_{\#}, l, (k-1 \text{ structure tier } X(A_{\#}), l(l)(l, l))$

Calculate X, Level down to (k-1)-th structure with tier X and Value X, nest the entire function into the (k-1)-th structure X-th tier bracket X times, with Innermost nested function =  $A_{\#}(k-1 \text{ Structure tier } X(X)), l(l)(l, l)$

## 7. Martin Extended Array Nunction(MEAN)

$L^v(k, n, l)$

Let v = array universe

Let k = k-th structure k-th tier value k operator. Such that  $l > 1$ .

Ignore k if  $l = 1$ .

Let n = value of each entries

Let l = length of p parameter

$L^1(n, n, 1) = A_{\#}(n)(n, n)$ , base Universe

$L^1(n, n, 2) = A_{\#}(n, (n\text{-th structure tier } n(n)), n)(n)(n, n)$

$L^2(1, n, 1) = A^1_{\#}(n)(n, n)$ , array universe 1

(note:

$A_{\#}$  = base Universe MAN

$A^n_{\#}$  = n-th universe MAN

$A(a, b, c\#)$  = base Universe MA

$A_n(a, b, c)$  = n-th universe MA)

General:

$L^n(n, n, n) = A^{n-1}_{\#}(n, (n\text{-th structure tier } n(n)), n)(n)(n, n)$

Calculating higher universe Martin Array Auction:

$A(a,b\#)$  = Martin Array Auction, Base Universe.

$A_1(a,b,\#)$  = Martin Array Auction, Universe 1.

$n(a,b,\#)$  = MAA level  $n$ , base Universe

$n_1(a,b,\#)$  = MAA level  $n$ , universe 1.

$A_1(a,b,c) = a \text{ (c-th hyperoperation level)}^b a$

Any  $v$  MA or MAA has the same mechanism as  $v = 1$ .

Leveling down universe:

$L^2(n,n,1) = A^1_{[n]}(n)(n,n) = \text{Let } X = \text{result of } A^1_{[n]}(n)(n,n).$

$\rightarrow L^1(X,X,X).$

## B.Rules

Rule 1: MAN array or function entry must contain an array, function, or non zero positive integers, except for operator slots.

Rule 2:  $(\#, \text{Operators}, k)$  or  $L_a(n, l, \text{operators}), b \backslash$  or  $(a, [n], \{n\}, /n, !, b)$  or  $L_a(n, l, \{n\}, /n, !)$ . Higher priority operator is written right before last entry, least Priority written after Second last array entry and so on, operator slots are always second to rightmost entry in MAN.

Rule 3: remove trailing 1,  $(\#, 1) \rightarrow (\#)$ , or  $(\#, j, !, 1) \rightarrow (\#, !, j)$ , or  $[\#1] \rightarrow [\#]$ , except for any first entry in MAN p or a parameters, except for any three entries in MA/MAA.

Rule 4: Prioritize  $(!)$  before  $m$ -th tier, then  $k$ -th structure.

Rule 5: Evaluate Innermost array/nested function first.

Rule 6:  $[n, /1/, /1/, n] \rightarrow [n, /2/, n]$

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